SELF-STUDY GUIDE

Databricks Free Edition

Taking the course on your own, step by step

Altkom Akademia
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This guide shows how to take the Databricks Fundamental course on your own in Databricks Free Edition. Free Edition costs nothing and needs no cloud subscription. The modules, both workshops and the final project are written to run there as well as on the classic cluster used in class. Tested status: pending. The full Free Edition path of this edition has not yet been run end to end; if a cell fails, check Part 7 first.

Item	Detail
What you need	A Databricks Free Edition account and a web browser
Setup time	About 10 minutes
Course time	370 minutes of teaching and workshops in the instructor-led day; working alone usually takes longer, and the BONUS sections add more
Compute	Serverless only – you never create a cluster
Course repository	`https://github.com/ditw-training/databricks-fundamental`

PART 1

Before you start

What Free Edition is

Databricks Free Edition is a free, non-commercial workspace for learning. It contains the same platform you would use at work: notebooks, Unity Catalog, Delta Lake, Lakeflow Jobs, the SQL Editor and AI/BI. You sign in with an e-mail one-time code or a Google or Microsoft account.

Limits that matter for this course

Limit	What it means for you
Serverless compute only	Notebooks and jobs always run on Serverless. A few BONUS demos behave differently (see Part 6)
One SQL warehouse, size 2X-Small	Enough for the SQL Editor, dashboards and Genie Agents in this course
At most 5 job tasks running at the same time	The course jobs have three tasks each – run one job at a time
One workspace, no account console	You cannot create account groups; the Unity Catalog demo uses a built-in group instead
No custom storage locations	Your catalog is stored in the workspace's built-in storage
Daily usage quota	If you exceed it, compute is switched off until the next day. Your data, notebooks and settings are kept
Non-commercial use only	Use the workspace for learning, not for company data

Databricks can change these limits; the current list is in the Databricks documentation under *Free Edition limitations*.

How the course is organised

The repository contains three kinds of notebooks:

Folder	Content	How to use it
`notebooks/setup`	Environment setup	Run `00_pre_config` once, then `00_setup` once
`notebooks/modules`	Demo modules M00 to M07	Read the explanations and run the cells one by one
`notebooks/workshops`	Hands-on exercises and the final project	Write your own code, then run the CHECK cell after each task

The data comes from the fictional online store RetailHub: customers, orders and products in CSV, JSON, Parquet and Excel files in the `dataset` folder.

PART 2

Set up the workspace

Step 1 – Create a Free Edition account

1. Open `https://www.databricks.com/learn/free-edition` and choose **Sign up**.
2. Sign in with your e-mail address, Google or Microsoft account.
3. Wait until the workspace opens. It already contains the catalogs `workspace`, `samples` and `system`.

Your course catalog is named after the account you sign in with, so use the same account for the whole course.

Step 2 – Clone the course repository as a Git folder

1. In the left menu open **Workspace**, then **Home** (your user folder).
2. Select **Create**, then **Git folder**.
3. Enter the Git repository URL `https://github.com/ditw-training/databricks-fundamental`. The provider GitHub is detected automatically. The repository is public, so no credentials are needed.
4. Select **Create Git folder**.

The folder `databricks-fundamental` appears in your Home folder with the subfolders `notebooks`, `dataset`, `materials`, `assets` and `docs`.

> Keep the notebooks inside the Git folder. The notebooks find the setup notebook and the data files through paths relative to the repository; a notebook copied elsewhere fails at `%run ../setup/00_setup`.

Step 3 – Run `00_pre_config`

1. Open `notebooks/setup/00_pre_config`.
2. The notebook is written for a trainer who prepares catalogs for a whole group. For Free Edition, run it for one person: in the configuration cell (Step 0) set `SINGLE_USER_MODE = True`. Leave the training group and the storage location as they are; single-user mode does not use them.

3. In the top-right corner select **Connect**, then **Serverless**.
4. Select **Run all**.

The notebook prints which mode it runs in and what it adjusted, for example that the catalog uses the built-in storage and which group the Unity Catalog module will use. The last cell reports whether the environment is ready. The notebook has created:

Object	Name
Catalog	`retailhub_<name>`
Schemas	`bronze`, `silver`, `gold`, `default`
Volume	`retailhub_<name>.default.datasets`, with a copy of the `dataset` folder
Permissions	`ALL PRIVILEGES` and `MANAGE` on the catalog for your account

Running `00_pre_config` again is safe: it creates only what is missing and copies the data files again.

Your catalog name

The catalog is named `retailhub_<name>`, where `<name>` is derived from your e-mail address. You do not need to work it out by hand: `00_setup` prints it (Step 4 and Step 5). Use the printed value wherever this guide or the notebooks say `retailhub_<name>`, for example in the job parameters and in the SQL Editor.

Step 4 – Run `00_setup`

Open `notebooks/setup/00_setup`, connect to Serverless and select **Run all**. The notebook checks the catalog, the schemas and the volume, and ends with a summary such as:

```
TRAINING ENVIRONMENT READY
  CATALOG:        retailhub_<name>
  DATASET_PATH:   /Volumes/retailhub_<name>/default/datasets
  IS_SERVERLESS:  True
```

Every module and workshop runs this notebook with `%run ../setup/00_setup` and receives the variables `CATALOG`, `BRONZE_SCHEMA`, `SILVER_SCHEMA`, `GOLD_SCHEMA`, `DATASET_PATH` and `IS_SERVERLESS`. You do not run it by hand again.

Step 5 – Check the environment

Open `notebooks/modules/01_platform_workspace` and run section 1, *Connect to your training catalog*. It runs `%run ../setup/00_setup` and prints your catalog name. Note the catalog name down; you need it in Part 5.

PART 3

Working with the notebooks

Compute

Always connect notebooks to **Serverless**. It starts in a few seconds and stops by itself when you are idle, so there is nothing to switch off. For the SQL Editor, dashboards and Genie Agents use the **Serverless Starter Warehouse**.

Running cells

Run the module cells one at a time and read the explanation above each cell; the course is written that way. **Run all** works too, but you learn less from it. You can re-run a module at any time: the demo cells recreate their tables.

Markers used in the notebooks

Marker	Meaning	What to do
⊕ **BONUS**	Outside the one-day schedule	Skip on the first pass, return to it later
⭐ **OPTIONAL**	Workshop task beyond the core path	Do it when you have time
[UI DEMO]	A step done in the Databricks interface, not in code	Follow the checklist in the cell
`%skip` at the top of a cell	The cell is not run by **Run all**	Remove the line when the text tells you to
`# YOUR CODE HERE`	A place for your code in a workshop	Replace it with your solution
CHECK cell	Automatic test of the previous task	Run it; all assertions must pass before you continue

Workshops

Each workshop task has a task cell and a CHECK cell. Write the code, run the task cell, then run the CHECK cell. A failed assertion tells you what is wrong. Fix the task cell and run both cells again.

Tasks marked ⭐ OPTIONAL go beyond the core path. If an optional task cell or its CHECK cell starts with `%skip`, delete that line in both cells to run them.

The workshops write to their own tables (`lab_ingestion_customers`, `lab_delta_cust_ws`, `lab_orders`, `lab_daily_orders`), so they never overwrite the demo tables.

Getting updates

When the course repository is updated, open the Git dialog (select **Git** next to the folder name, or the branch button `main` in a notebook) and select **Pull**.

If Pull reports a conflict, you changed a notebook that was also updated in the repository. Copy your version outside the Git folder (**Clone** in the notebook menu), then in the Git dialog select ⋮, **Reset** to discard local changes, and **Pull** again.

PART 4

The course step by step

The order below follows the one-day schedule. Times are those of the instructor-led day; working alone usually takes longer. In each module, everything after the Summary, under a ⊕ BONUS banner, is extra.

M00 – Intro and setup (15 min)

No notebook: the opening slides cover the objectives, the agenda and the RetailHub data. The environment check is Part 2, Step 5.

M01 – Platform, costs and the interface (60 min)

`notebooks/modules/01_platform_workspace`. The workspace, Catalog Explorer, the three-level namespace, volumes, compute types and their costs, magic commands, Genie Code, and charts and data profiles without code.

Things to do by hand:

- **Upload files to a volume.** Download `stores.csv` and `promotions.json` from the repository folder `materials/volume_upload` on GitHub, then upload them in **Catalog** to the volume named in the notebook (**Upload to this volume**). Or run the copy cell in the notebook, which copies them from your Git folder. Run the next cells to read the files.
- **Visualisations and data profile.** After a `display()` result select **+**, then **Visualization** or **Data Profile**.

Free Edition notes: the billing query on `system.billing.usage` is a trainer demo. If it fails with a permission or not-found error in Free Edition, skip it. On Serverless, Photon is always on.

M02 – How Spark works (15 min)

`notebooks/modules/02_spark_architecture`. Driver and executors, lazy evaluation and one engine for SQL and PySpark, on the `samples.nyctaxi.trips` table. The BONUS covers stages, shuffles, AQE, the Spark UI and cache.

Free Edition notes:

- Serverless lets you read only some Spark settings, so the BONUS AQE cell shows the defaults for the others.
- DataFrame caching on Serverless is in Public Preview, introduced in October 2026. It needs serverless environment version 5 or above, it is on by default and a workspace admin can switch it off. On Serverless, `cache()` computes the result at once and keeps it as a snapshot for your session; SQL `CACHE TABLE` is not supported. If caching is not available, the BONUS cell prints a message instead of failing.
- There is no Spark UI. Use the query profile instead: **See performance** under the cell result, then **See query profile**.

M03 – Medallion and ingestion (50 min)

`notebooks/modules/03_medallion_ingestion`. A five-minute introduction to Python, the medallion architecture, CSV and JSON with an explicit schema, `read_files()`, exploring data, transformations with their SQL equivalents, temporary views and writing to a Delta table. Parquet and Excel are BONUS.

Free Edition notes: Excel files are read with the built-in Excel reader, which needs Databricks Runtime 17.1 or above on classic compute. The BONUS *Global Temporary Views* is skipped — Serverless does not support global temporary views. Regular temporary views work.

Workshop – Ingestion (40 min)

`notebooks/workshops/WORKSHOP_ingestion`. The CORE tasks go from reading CSV and JSON with a schema, through new columns, type casts, conditional logic and aggregations, to writing a Bronze Delta table with load metadata. The remaining tasks are ⭐ OPTIONAL; the table *Tasks at a Glance* in the notebook lists both groups.

M04 – Delta Lake (40 min)

`notebooks/modules/04_delta_fundamentals`. Creating a table, schema enforcement and evolution, INSERT, UPDATE, DELETE, MERGE, table history, time travel and RESTORE. VACUUM, UNDROP, liquid clustering and predictive optimization are explained as concepts.

Free Edition notes: the BONUS VACUUM demo shortens the retention of its own demo table and resets it afterwards. After VACUUM the old versions can no longer be read, which is the point of the demo. The BONUS *External Tables* reports that no external location is available — this is expected in Free Edition.

Workshop – Delta Lake (35 min)

`notebooks/workshops/WORKSHOP_delta`. CORE tasks 01, 03, 05 and 06, 08, 10 and 12: create a Delta table, add a column with schema evolution, change data with INSERT, UPDATE and MERGE, read an older version with time travel and recover the table with RESTORE. Tasks 02, 04, 07, 09 and 11 (schema enforcement, a CHECK constraint, DELETE, the history, time travel by timestamp) are ⭐ OPTIONAL.

M05 – Lakeflow Jobs (30 min)

`notebooks/modules/05_orchestration_jobs`. Lakeflow Jobs (formerly Workflows): compute for jobs, a three-task job built from ready notebooks, job and task parameters, task values, a schedule and a repair run. Lakeflow Spark Declarative Pipelines and job monitoring with system tables are BONUS. Build the job described in Part 5.

M06 – Unity Catalog (15 min)

`notebooks/modules/06_unity_catalog`. Account groups, then `GRANT`, `SHOW GRANTS` and `REVOKE`, and the same in the **Permissions** tab of Catalog Explorer.

Free Edition notes: there is no account console, so the group `retailhub_analysts` cannot exist. Follow the single-user note in the notebook: in single-user mode the demo grants to the built-in group `account users` instead, and `SHOW GRANTS` and `REVOKE` work the same way. In Free Edition you are the only member of `account users`.

Final project (60 min)

`notebooks/workshops/final_labs/WORKSHOP_final_project`. Build the orders pipeline from raw JSON to a daily revenue table and run it as a scheduled job.



1. Run `tasks/lab_task_01_bronze` — it is ready.
2. Complete and run `tasks/lab_task_02_silver` and `tasks/lab_task_03_gold`; run the CHECK cells.
3. Follow the steps in the project notebook: the smoke test, the job described in Part 5 and the SQL checks.

Expected results:

Table	Rows
`bronze.lab_orders`	100,000
`silver.lab_orders`	90,904 (invalid rows are removed)
`gold.lab_daily_orders`	32 (one row per day, 2024-12-31 to 2025-01-31)

M07 – Genie Agents and dashboards (optional)

`notebooks/modules/07_aibi_dashboards`. Run the notebook cells first: they create the demo tables in your Gold schema. Then follow the steps in the notebook in the interface: a Genie Agent (formerly Genie space) on the Gold tables, and an AI/BI dashboard. Use the Serverless Starter Warehouse.

PART 5

Jobs in Free Edition

Jobs are created in **Jobs & Pipelines**, then **Create**, **Job**. In Free Edition the only compute for a job task is **Serverless**. Wherever the notebooks say *Shared training cluster* or *Job cluster*, choose Serverless.

The M05 demo job

The three task notebooks are ready in `materials/orchestration`.

Task name	Notebook	Depends on
`validate`	`materials/orchestration/task_01_validate`	–
`transform`	`materials/orchestration/task_02_transform`	`validate`
`report`	`materials/orchestration/task_03_report`	`transform`

1. Create the job and name it, for example `Demo_ETL_Pipeline_<name>`.
2. Add the three tasks from the table. For each: type **Notebook**, source **Workspace**, compute **Serverless**.
3. Add the job parameters `source_table` = `samples.nyctaxi.trips` and `min_rows` = `100`. Job parameters reach every task; if a task parameter has the same key, the job parameter wins.
4. Add a trigger: **Scheduled**, **Advanced**, cron `0 0 6 * * ?`, timezone `Europe/Warsaw`, status **Paused**.
5. Select **Run now**, open the run and look at the task graph and the output of each task.
6. Select **Run now with different settings**, set `min_rows` = `999999999`: `validate` fails. Use **Repair run** with `min_rows` = `100`; only the failed task and the tasks after it run again.

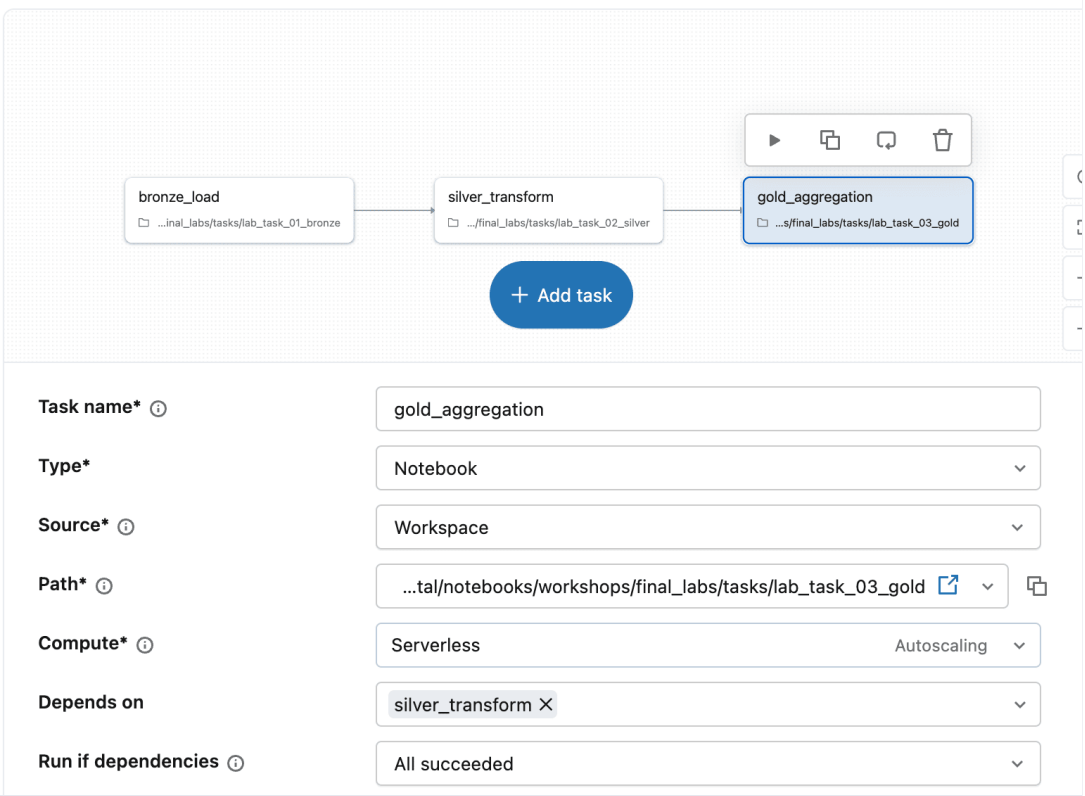
> The task names matter. `transform` reads a task value from the task named `validate`, and `report` reads from `transform`. With other names, the downstream task fails. Rename the task and use **Repair run**.

The final project job

Task name	Notebook	Depends on
`bronze_load`	`notebooks/workshops/final_labs/tasks/lab_task_01_bronze`	–
`silver_transform`	`notebooks/workshops/final_labs/tasks/lab_task_02_silver`	`bronze_load`
`gold_aggregation`	`notebooks/workshops/final_labs/tasks/lab_task_03_gold`	`silver_transform`

Job parameters, with the catalog name printed by `00_setup`:

Parameter	Value
`catalog`	`retailhub_<name>`
`source_path`	`/Volumes/retailhub_<name>/default/datasets`



Add the same paused schedule as for the M05 job and select **Run now**. Each task shows its result in the **Output** tab, for example the table it wrote and its row count. If a task fails, fix the notebook and use **Repair run**: only the failed task and the tasks after it run again.

Checking results in the SQL Editor

Open **SQL Editor**, select the **Serverless Starter Warehouse** and run:

```
SELECT * FROM retailhub_<name>.gold.lab_daily_orders ORDER BY order_date;

DESCRIBE HISTORY retailhub_<name>.gold.lab_daily_orders;
```

Keeping within the limits

- Run one job at a time: Free Edition allows five job tasks at once.
- Leave schedules **Paused**. A daily schedule keeps using your quota after you finish the course.

PART 6

Differences from the instructor-led course

The instructor-led course runs on Azure Databricks with a shared classic cluster. The notebooks detect Serverless and handle the known differences; the affected cells print an explanation instead of failing.

Module	Cell	In Free Edition
M01	Billing demo (trainer)	Reads `system.billing.usage`; if it fails with a permission or not-found error, skip it
M01	BONUS Cluster Info and Monitoring	Runtime `client.x.y`; Photon is always on
M02	BONUS AQE settings	Only some Spark settings can be read; defaults are shown for the others
M02	BONUS Cache and Persist	Public Preview on Serverless (environment version 5 or above): computed at once as a session snapshot; otherwise the cell prints a message
M02	BONUS Spark UI	No Spark UI; use the query profile of a cell
M03	BONUS Global Temporary Views	Not supported on Serverless; skipped
M04	BONUS VACUUM	The demo shortens the retention of its demo table, then resets it
M04	BONUS External Tables	No external location; the cell reports it
M06	GRANT / REVOKE	Single-user mode: grants to `account users` instead of `retailhub_analysts`
M05, Final project	Job compute	Serverless instead of a cluster

Reading files from volumes, Delta Lake, MERGE, time travel, RESTORE, jobs with parameters and task values and the SQL Editor are expected to work the same way. Tested status: pending, until this edition of the course has been run end to end in Free Edition.

PART 7

Troubleshooting

Symptom	Cause	Fix
`Catalog 'retailhub_<name>' not found`	`00_pre_config` has not been run, or you use a different account	Run `00_pre_config` with the account you use for the course
`%run ../setup/00_setup` fails with *notebook not found*	The notebook was copied outside the Git folder	Open the notebook from `databricks-fundamental`
The volume `datasets` is empty	The data copy in `00_pre_config` did not finish	Run `00_pre_config` again
`CONFIG_NOT_AVAILABLE` or `NOT_SUPPORTED_WITH_SERVERLESS`	A feature or setting that Serverless does not support, often in an older copy of the repository	Pull the latest version (Part 3, *Getting updates*); if the cell is BONUS, skip it
A CHECK cell fails	The task is not finished or a variable has a different name	Read the assertion message, fix the task cell, run both cells again
`TABLE_OR_VIEW_NOT_FOUND` for `lab_orders` in Silver or Gold	The previous layer has not been built	Run the Bronze, then the Silver task first
The `transform` or `report` task of the M05 job fails	The tasks have other names than `validate` and `transform`	Rename the tasks (Part 5), then **Repair run**
Compute is unavailable for the rest of the day	The daily usage quota was exceeded	Wait until the next day; data and notebooks are kept
Pull reports a conflict	You edited a notebook that changed in the repository	Clone your notebook elsewhere, then **Reset** and **Pull**

PART 8

Cleaning up

When you have finished the course, remove what it created. To start again later, run `00_pre_config` and `00_setup` again.

1. In **Jobs & Pipelines** delete the M05 job and the final project job, or at least leave their schedules paused.
2. Delete the Genie Agent and the dashboard from M07, if you created them.
3. Drop the course catalog in a notebook or in the SQL Editor:

```
DROP CATALOG IF EXISTS retailhub_<name> CASCADE;
```

4. Optionally delete the Git folder `databricks-fundamental`.